THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS UNIT TEST SET C

Full Syllabus (Chapters 9–14) | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Which pair of statements are converses of each other and both true? 1

(a) "If it rains, the road is wet" and "If the road is wet, it has rained"

(b) "If a number ends in 0, then it is even" and "If a number is even, then it ends in 0"

(c) "If a triangle is equilateral, then all its angles are equal" and "If all angles of a triangle are equal, then it is equilateral"

(d) "If $x = 2$, then $x^2 = 4$" and "If $x^2 = 4$, then $x = 2$"

2. A triangle has sides 9 cm, 12 cm and 16 cm. By the converse of the Baudhāyana–Pythagoras theorem it is: 1

(a) right-angled, since $a^2 + b^2 = c^2$

(b) not right-angled, since $a^2 + b^2 \neq c^2$

(c) right-angled, since $a + b > c$

(d) right-angled, since the sides are in increasing order

3. For two groups of equal size, the average of the combined group equals: 1

(a) the simple average of the two group averages

(b) the smaller average

(c) the larger average

(d) their product

4. A group of 24 students has an average of 68 marks and another group of 16 students has an average of 78 marks. The average of all 40 students is: 1

(a) 73 **(b)** 72 **(c)** 74 **(d)** 71

5. By Euclid's algorithm, gcd(252, 198) is: 1

(a) 36 **(b)** 18 **(c)** 9 **(d)** 19

6. When we move from adding 3-digit numbers to adding 5-digit numbers, which statement is correct? 1

(a) counting dots needs 2 times as many dots and the column method 100 times as many steps

(b) counting dots needs about 100 times as many dots, but the column method needs only 2 more columns

(c) both methods need about 100 times as much work

(d) both methods need only 2 more steps

7. The figure that is a non-convex quadrilateral is the one that: 1

(a) has all angles equal **(b)** has one internal angle greater than 180°

(c) has no diagonals **(d)** has two parallel sides

8. **Assertion (A):** In a triangle ABC the segment PQ joining the midpoints of AB and AC has length BC/2. 1

Reason (R): PQ is parallel to BC.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

9. The number of solutions of a linear equation in two variables is: 1

(a) zero **(b)** exactly two **(c)** infinite **(d)** exactly one

10. The point (1, k) lies on the line 3x + 2y = 7. The value of k is: 1

(a) 2 **(b)** 1 **(c)** 4 **(d)** 3

11. The volume of a cuboid measuring 12 cm $\times$ 10 cm $\times$ 5 cm is: 1

(a) 27 cm^3 **(b)** 620 cm^3 **(c)** 460 cm^3 **(d)** 600 cm^3

12. Taking $\pi = \frac{22}{7}$, the curved surface area of a cylinder of radius 14 cm and height 6 cm is: 1

(a) 550 cm^2 **(b)** 264 cm^2 **(c)** 528 cm^2 **(d)** 3696 cm^2

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

13. State the converse of "If a number is divisible by 18, then it is divisible by 9." Is it true? Justify. 2

14. The average marks of 30 girls are 70 and of 20 boys are 60. Find the average of all 50 students. 2

15. State the 'equal parallel sides' test for a parallelogram. 2

16. State the slope of the lines 2

(i) $y = -3$ and

(ii) $x = 2$.

17. How many cubes of side 2 cm can be cut from a cuboid of 10 cm $\times$ 8 cm $\times$ 6 cm? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Show that the number 143 gives a counterexample to the claim "$n^2 + n + 11$ is prime for every natural number n". State the value of n used. 3

19. A bird flew an average of 40 km per day for 20 days and then 61 km on the 21st day. Make a guess, then find the 21-day average. 3

20. A cab company charges a fixed fare plus a fixed rate per km. For 8 km the fare is ₹140 and for 12 km it is ₹200. Find the fixed fare, the rate per km and the fare for 20 km. 3

21. The curved surface area of a cone is 462 cm^2 and its slant height is 21 cm. Find the base radius and the total surface area ($\pi = \frac{22}{7}$). 3

22. A cylindrical drum has radius 0.7 m and height 1.5 m. Find the volume of oil it can hold in litres ($\pi = \frac{22}{7}$, 1 m$^3 = 1000$ litres). 3

OR

The radii of two cylinders are in the ratio $1 : 2$ and their heights are in the ratio $4 : 3$. Find the ratio of 3

(i) their volumes,

(ii) their curved surface areas.

23. A class says "If it is Monday, our school has an assembly." Write the converse and explain why it may not be true. 3

OR

Test whether the triangle with sides 12 cm, 35 cm, 37 cm is right-angled. Name the theorem you use. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. (i) Prove that the combined average of collections of n and m values with averages a and b is $\frac{an+bm}{n+m}$. 5

(ii) Hence show that when $n = m$ it reduces to $\frac{a+b}{2}$.

OR

A zoo has 50 deer with average weight 40 kg; males average 46 kg and females 36 kg. 5

(i) Form the equation for the number of males x.

(ii) Find the number of males and females.

(iii) A female deer of 42 kg joins the zoo. Find the new average weight of the females.

25. Prove the Centroid Theorem: the medians of a triangle are concurrent and divide each other in the ratio 2 : 1. (Use midpoints of CM and BM.) 5

OR

In quadrilateral ABCD, P, Q, R, S are the midpoints of AB, BC, CD, DA. Show that PR and QS bisect each other and that if AC $=$ BD then PR $\perp$ QS. 5

END OF QUESTION PAPER — **Total: 50 Marks**